MAHESH VEERABOINA

Physical Design Engineer | Hyderabad, Telangana, India
University College of Engineering, Osmania University

PROFILE SUMMARY

Physical Design Engineer working on 28nm RTL-to-GDSII implementation. I've taken blocks from floorplan through DRC/LVS signoff - the largest being a 350K-cell SoC block with 24 macros and 5 clock domains. Comfortable automating PnR flows with TCL and Makefiles on Cadence Innovus/Genus, and currently building depth in low-power design and power integrity. Looking to apply this in high-performance mixed-signal IP work.

WORK EXPERIENCE

Physical Design Engineer - SemiconOS May 2026 - Present

- Went through intensive hands-on training on the full RTL-to-GDSII flow (Cadence Innovus + Genus, 28nm): floorplanning, power grid, placement, CTS, routing, timing closure, DRC fixes.
- Independently completed three block-level projects - RPTOP (69K gates), ALU (428 gates), and Tile (302K cells, 18 macros) - each delivered as DRC-clean GDSII.
- Scripted the repetitive parts (constraint loading, placement runs, post-route reports) in TCL/TCSH so each iteration takes less manual work.

Physical Design Engineer Intern - RiseTime Semiconductors, Hyderabad June 2025 - May 2026

- Automated the end-to-end PnR flow (floorplan to signoff) with TCL scripts and Makefiles, so runs are repeatable across design blocks instead of being redone by hand each time.
- Owned the physical implementation of the Iguana SoC block - ~350K standard cells, 24 hard macros, 5 independent clock domains - covering floorplan architecture, macro placement, multi-clock CTS, timing convergence, and congestion cleanup.
- Wrote reusable TCL utilities for congestion analysis, timing report parsing, and design-rule checks - these cut down iteration time noticeably during closure.
- Worked with senior engineers on power grid strategy and placement guidelines for blocks with multiple power domains and tight PPA targets.
- Handled DRC/LVS signoff-readiness checks for large block implementations.

TECHNICAL SKILLS

Physical Design Flow: Floorplanning, Power Planning & Power Grid Design, Placement, Clock Tree Synthesis (CTS), Routing, Timing Closure, DRC/LVS Signoff - full RTL-to-GDSII on 28nm

EDA Tools: Cadence Innovus (Place & Route), Cadence Genus (Synthesis); basic exposure to Synopsys tools and formal equivalence verification (Formality)

Timing & Verification: Static Timing Analysis (STA), setup/hold violation fixing, clock domain analysis, physical verification (DRC, LVS, antenna checks)

Design Knowledge: CMOS fundamentals, PPA trade-offs, multi-clock domain design, hard macro placement, congestion resolution, low-power concepts and multiple power domain awareness

Scripting & Automation: TCL, TCSH, Makefiles - flow automation, report generation, constraint management, DRC checking scripts

HDL: Basic Verilog HDL

PROJECTS

Iguana - SoC Block-Level Physical Implementation

28nm Physical Design (Place & Route) | Cadence Innovus

- The most complex block I've worked on: ~350K cells, 24 hard macros, 5 clock domains - the floorplan and power grid had to account for all five domains from day one.
- Responsible for macro placement, floorplan definition, and multi-clock CTS to meet skew and insertion-delay targets across all domains.
- Built the Makefile/TCL automation that runs the full flow reproducibly; currently driving timing convergence and congestion fixes in high-utilization regions ahead of DRC/LVS signoff.

Tile - Block-Level Physical Implementation

28nm RTL to GDSII | Cadence Innovus

- Largest block I've closed independently: 302,511 cells, 18 hard macros, 1.7ns clock.
- Ran the full flow - hierarchical floorplan, macro placement, multi-layer power grid, congestion-driven placement, CTS with skew balancing, detailed routing - through to DRC-clean GDSII.
- Closed timing at 1.7ns by working through setup/hold violations with placement optimization and clock-tree tuning; automated iterative runs with TCL/Makefiles to keep turnaround short.

RPTOP - Block-Level Implementation

28nm RTL to GDSII | Cadence Innovus & Genus

- 69K-gate block, 6 hard macros, one 1.8ns primary clock plus a virtual clock - did the floorplan, power rings/straps, CTS, and detailed routing.
- Fixed routing congestion by tuning placement constraints, cleared all setup/hold violations, and delivered DRC-clean GDSII; constraint loading and reporting scripted in TCL.

ALU - Logic Synthesis

28nm RTL to Gate-Level Netlist | Cadence Genus

- Synthesized a 428-gate ALU with primary (2ns), generated, and virtual clocks; applied SDC constraints and optimized for timing and area.
- Verified QoR with Genus reports and confirmed formal equivalence of the synthesized netlist.

EDUCATION

B.E. in Electrical and Electronics Engineering - University College of Engineering, Osmania University — May 2025 | 78%

Intermediate (MPC) - Narayana Junior College, Hyderabad — March 2021 | 97%

ACADEMIC ACHIEVEMENTS

- Awarded the Foundation for Excellence Scholarship - a merit-based national scholarship for academic excellence (2021-2022).